MEDIUM ALGEBRA

Linear functions 04

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

Energy per Gram of Typical Macronutrients

Macronutrient	Food calories	Kilojoules
Protein	4.0	16.7
Fat	9.0	37.7
Carbohydrate	4.0	16.7

The table above gives the typical amounts of energy per gram, expressed in both food calories and kilojoules, of the three macronutrients in food. If x food calories is equivalent to k kilojoules, of the following, which best represents the relationship between x and k ?

A. $k = 0.24x$

B. $k = 4.2x$

C. $x = 4.2k$

D. $xk = 4.2$

Q2 GRID-IN ALGEBRA

The function f is defined by $f(x) = mx + b$, where m and b are constants. If $f(0) = 18$ and $f(1) = 20$, what is the value of m ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ALGEBRA

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is $\$ 220$ when it sells 8 items, and its profit is $\$ 320$ when it sells 10 items. Which equation defines f?

A. $f x = 150x - 320$

B. $f x = 32x$

C. $f x = 50x - 10x$

D. $f x = 50x - 180$

Q4 ALGEBRA

For the linear function g, the graph of $y = g(x)$ in the xy-plane has a slope of 2 and passes through the point $1, 14$. Which equation defines g?

A. $g(x) = 2x$

B. $g(x) = 2x + 2$

C. $g(x) = 2x + 12$

D. $g(x) = 2x + 14$

Q5 ALGEBRA

Number of cars	Maximum number of passengers and crew
3	174
5	284
10	559

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A. $55c - p = -9$

B. $55c - p = 9$

C. $55p - c = -9$

D. $55p - c = 9$

Q6

ALGEBRA

$$f(x) = 39$$

For the given linear function f, which table gives three values of x and their corresponding values of fx?

A.

x	fx
0	0
1	0
2	0

B.

x	fx
0	39
1	39
2	39

C.

x	fx
0	0
1	39
2	78

D.

x	fx
0	39
1	0
2	-39

Q7

ALGEBRA

In the xy-plane, the graph of the linear function f contains the points $0, 2$ and $8, 34$. Which equation defines f, where $y = fx$?

A. $fx = 2x + 42$

B. $fx = 32x + 36$

C. $fx = 4x + 2$

D. $fx = 8x + 2$

Q8

GRID-IN ALGEBRA

The function f is defined by $f x = \text{-}9x + 9$. What is the y-coordinate of the y-intercept of the graph of $y = f x$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 ALGEBRA

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers. Which equation could represent this linear relationship?

A. $t = 4d$

B. $t = \frac{1}{25}d$

C. $t = 25d$

D. $t = \frac{1}{4}d$

x	gx
1	54
2	51
3	48
4	45

For the linear function g, the table shows four values of x and their corresponding values of gx. The function can be written as $gx = mx + b$, where m and b are constants. What is the value of b?

A. 3

B. 27

C. 54

D. 57